Chapter 13: Derivative Rules (Quy tắc Tính Đạo hàm)

📖 Chapter Overview

Computing derivatives using the limit definition can be tedious and time-consuming. Fortunately, mathematicians have developed efficient rules and formulas that allow us to find derivatives quickly and systematically. In this chapter, we explore the fundamental rules of differentiation: the power rule, sum and difference rules, product and quotient rules, and the chain rule. We'll also learn the derivatives of common functions including polynomials, exponential, logarithmic, and trigonometric functions. These rules are the essential tools that make calculus practical for solving real-world problems in physics, engineering, economics, and beyond.

📚 Bilingual Vocabulary (Từ vựng Song ngữ)

Basic Rules (Quy tắc Cơ bản)

English	IPA	Vietnamese	Example
Power rule	/ˈpaʊər ruːl/	Quy tắc lũy thừa	$(x^n)' = nx^{n-1}$
Constant rule	/ˈkɑːnstənt ruːl/	Quy tắc hằng số	$(c)' = 0$
Constant multiple rule	/ˈkɑːnstənt ˈmʌltɪpəl ruːl/	Quy tắc nhân hằng số	$(cf)' = cf'$
Sum rule	/sʌm ruːl/	Quy tắc tổng	$(f + g)' = f' + g'$
Difference rule	/ˈdɪfərəns ruːl/	Quy tắc hiệu	$(f - g)' = f' - g'$

Advanced Rules (Quy tắc Nâng cao)

English	IPA	Vietnamese	Example
Product rule	/ˈprɑːdʌkt ruːl/	Quy tắc tích	$(fg)' = f'g + fg'$

Quotient rule	/ˈkwoʊʃənt ruːl/	Quy tắc thương	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
Chain rule	/tʃeɪn ruːl/	Quy tắc hàm hợp	$(f \circ g)' = f'(g) \cdot g'$
Composite function	/ˈkɑːmpəzɪt ˈfʌŋkʃən/	Hàm hợp	$y = f(g(x))$
Outer function	/ˈaʊtər ˈfʌŋkʃən/	Hàm ngoài	In $f(g(x))$, f is the outer function.
Inner function	/ˈɪnər ˈfʌŋkʃən/	Hàm trong	In $f(g(x))$, g is the inner function.

Function Types (Loại Hàm số)

English	IPA	Vietnamese	Example
Polynomial function	/ˌpɑːliˈnoʊmiəl ˈfʌŋkʃən/	Hàm đa thức	$f(x) = 3x^4 - 2x^2 + 5$
Rational function	/ˈræʃənəl ˈfʌŋkʃən/	Hàm phân thức	$f(x) = \frac{x+1}{x-2}$
Exponential function	/ˌekspəˈnenʃəl ˈfʌŋkʃən/	Hàm mũ	$f(x) = e^x$
Logarithmic function	/ˌlɔːgəˈrɪðmɪk ˈfʌŋkʃən/	Hàm logarit	$f(x) = \ln x$

Trigonometric function	/ˌtrɪgənəˈmetrɪk ˈfʌŋkʃən/	Hàm lượng giác	$f(x) = \sin x$

13.1 Basic Derivative Rules (Quy tắc Cơ bản)

Constant Rule

> **Theorem:** The derivative of a constant is zero.
> $\frac{d}{dx}(c) = 0$

Explanation: A constant function has a horizontal graph with slope 0, so its derivative is 0 everywhere.

Examples:

- $\frac{d}{dx}(5) = 0$
- $\frac{d}{dx}(\pi) = 0$
- $\frac{d}{dx}(-100) = 0$

Power Rule

> **Theorem:** For any real number
> n
> ,
> $\frac{d}{dx}(x^n) = nx^{n-1}$

This is one of the most important and frequently used differentiation rules.

Examples:

- $\frac{d}{dx}(x) = 1 \cdot x^0 = 1$

- $\frac{d}{dx}(x^2) = 2x$
- $\frac{d}{dx}(x^3) = 3x^2$
- $\frac{d}{dx}(x^{10}) = 10x^9$
- $\frac{d}{dx}(\sqrt{x}) = \frac{d}{dx}(x^{1/2}) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$
- $\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}(x^{-1}) = -x^{-2} = -\frac{1}{x^2}$

Constant Multiple Rule

> **Theorem:** If
> c
> is a constant and
> f
> is a differentiable function, then
> $\frac{d}{dx}[cf(x)] = c \cdot f'(x)$

Explanation: Constants can be "pulled out" of the derivative.

Examples:

- $\frac{d}{dx}(3x^2) = 3 \cdot \frac{d}{dx}(x^2) = 3 \cdot 2x = 6x$
- $\frac{d}{dx}(-5x^4) = -5 \cdot 4x^3 = -20x^3$
- $\frac{d}{dx}(2\sqrt{x}) = 2 \cdot \frac{1}{2\sqrt{x}} = \frac{1}{\sqrt{x}}$

Sum and Difference Rules

> **Theorem:** If
> f
> and

> g
> are differentiable functions, then
>
> - **Sum Rule:**
> $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$
>
> - **Difference Rule:**
> $\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$

Explanation: The derivative of a sum (or difference) is the sum (or difference) of the derivatives.

Examples:

- $\frac{d}{dx}(x^3 + x^2) = 3x^2 + 2x$
- $\frac{d}{dx}(5x^4 - 3x^2 + 7) = 20x^3 - 6x$
- $\frac{d}{dx}(x^2 + \sqrt{x} - \frac{1}{x}) = 2x + \frac{1}{2\sqrt{x}} + \frac{1}{x^2}$

13.2 Product and Quotient Rules (Quy tắc Tích và Thương)

Product Rule

> **Theorem:** If
> f
> and
> g
> are differentiable functions, then
> $\frac{d}{dx}[f(x) \cdot g(x)] = f'(x) \cdot g(x) + f(x) \cdot g'(x)$
>
> In words: The derivative of a product is the derivative of the first times the second, plus the first times the derivative of the second.

Memory aid:
$(fg)' = f'g + fg'$

(first times second's derivative plus second times first's derivative)

Important:
$(fg)' eq f' g'$
— This is a common mistake!

Examples:

1. $\frac{d}{dx}[(x^2)(x^3)] = (2x)(x^3) + (x^2)(3x^2) = 2x^4 + 3x^4 = 5x^4$

   (Note: We could also simplify first:
   $x^2 \cdot x^3 = x^5$
   , so
   $\frac{d}{dx}(x^5) = 5x^4$
)
2. $\frac{d}{dx}[(x^2 + 1)(x^3 - 2x)]$

   Let
   $f(x) = x^2 + 1$
   and
   $g(x) = x^3 - 2x$

   $f'(x) = 2x$
   and
   $g'(x) = 3x^2 - 2$

   $(fg)' = (2x)(x^3 - 2x) + (x^2 + 1)(3x^2 - 2)$
   $= 2x^4 - 4x^2 + 3x^4 - 2x^2 + 3x^2 - 2$
   $= 5x^4 - 3x^2 - 2$

Quotient Rule

> **Theorem:** If
> f
> and
> g
> are differentiable functions and

> $g(x) eq 0$
> , then
> $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)\cdot g(x) - f(x)\cdot g'(x)}{[g(x)]^2}$
>
> In words: The derivative of a quotient is (derivative of top times bottom minus top times derivative of bottom) all over bottom squared.

Memory aid:
$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
— "lo d-hi minus hi d-lo over lo-lo"

Important: The order matters! It's
$f'g - fg'$
, not
$fg' - f'g$
.

Examples:

1. $\frac{d}{dx}\left(\frac{x^2}{x+1}\right)$

   Let
   $f(x) = x^2$
   and
   $g(x) = x + 1$

   $f'(x) = 2x$
   and
   $g'(x) = 1$

   $\frac{d}{dx}\left(\frac{x^2}{x+1}\right) = \frac{(2x)(x+1) - (x^2)(1)}{(x+1)^2} = \frac{2x^2 + 2x - x^2}{(x+1)^2} = \frac{x^2 + 2x}{(x+1)^2}$

2. $\frac{d}{dx}\left(\frac{x-1}{2x+1}\right)$

   $= \frac{(1)(2x+1) - (x-1)(2)}{(2x+1)^2} = \frac{2x+1-2x+2}{(2x+1)^2} = \frac{3}{(2x+1)^2}$

13.3 The Chain Rule (Quy tắc Hàm hợp)

Composite Functions

A **composite function** is a function formed by applying one function to the result of another.

If
$y = f(u)$
and
$u = g(x)$
, then
$y = f(g(x))$
is a composite function.

Examples:

- $y = (x^2 + 1)^3$
  — outer:
  u^3
  , inner:
  $u = x^2 + 1$

- $y = \sqrt{x^2 + 5}$
  — outer:
  $\sqrt{u}$
  , inner:
  $u = x^2 + 5$

- $y = e^{x^2}$
  — outer:
  e^u
  , inner:
  $u = x^2$

The Chain Rule

> **Theorem:** If
> g

> is differentiable at
> x
> and
> f
> is differentiable at
> $g(x)$
> , then the composite function
> $F = f \circ g$
> defined by
> $F(x) = f(g(x))$
> is differentiable at
> x
> , and
> $F'(x) = f'(g(x)) \cdot g'(x)$
>
> In Leibniz notation: If
> $y = f(u)$
> and
> $u = g(x)$
> , then
> $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

In words: The derivative of a composite function is the derivative of the outer function (evaluated at the inner function) times the derivative of the inner function.

Strategy:

1. Identify the outer function
   f
   and inner function
   g

2. Find
   $f'(u)$
   where
   $u = g(x)$

3. Find
   $g'(x)$

4. Multiply:
   $f'(g(x)) \cdot g'(x)$

Examples:

1. $\frac{d}{dx}[(x^2 + 1)^3]$

   Outer:
   $f(u) = u^3$
   , so
   $f'(u) = 3u^2$

   Inner:
   $u = g(x) = x^2 + 1$
   , so
   $g'(x) = 2x$

   $\frac{d}{dx}[(x^2 + 1)^3] = 3(x^2 + 1)^2 \cdot 2x = 6x(x^2 + 1)^2$

2. $\frac{d}{dx}[\sqrt{x^2 + 5}]$

   Outer:
   $f(u) = \sqrt{u} = u^{1/2}$
   , so
   $f'(u) = \frac{1}{2}u^{-1/2} = \frac{1}{2\sqrt{u}}$

   Inner:
   $u = x^2 + 5$
   , so
   $g'(x) = 2x$

   $\frac{d}{dx}[\sqrt{x^2 + 5}] = \frac{1}{2\sqrt{x^2+5}} \cdot 2x = \frac{x}{\sqrt{x^2+5}}$

3. $\frac{d}{dx}[(3x-1)^{10}]$

$$= 10(3x-1)^9 \cdot 3 = 30(3x-1)^9$$

13.4 Derivatives of Special Functions (Đạo hàm các Hàm đặc biệt)

Exponential Functions

> **Formulas:**
> - $\frac{d}{dx}(e^x) = e^x$
> - $\frac{d}{dx}(a^x) = a^x \ln a$
>   (for
> $a > 0$
>   ,
> $aeq1$
>   )

The function
e^x
is unique: it equals its own derivative!

With chain rule:

- $\frac{d}{dx}(e^u) = e^u \cdot u'$
- $\frac{d}{dx}(a^u) = a^u \ln a \cdot u'$

Examples:

- $\frac{d}{dx}(e^{x^2}) = e^{x^2} \cdot 2x = 2xe^{x^2}$
- $\frac{d}{dx}(e^{3x}) = e^{3x} \cdot 3 = 3e^{3x}$

- $\frac{d}{dx}(2^x) = 2^x \ln 2$

Logarithmic Functions

> **Formulas:**
>
> - $\frac{d}{dx}(\ln x) = \frac{1}{x}$
>   (for
> $x > 0$
>   )
> - $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$
>   (for
> $x > 0$
>   ,
> $a > 0$
>   ,
> $aeq1$
>   )

With chain rule:

- $\frac{d}{dx}(\ln u) = \frac{1}{u} \cdot u' = \frac{u'}{u}$
- $\frac{d}{dx}(\log_a u) = \frac{1}{u \ln a} \cdot u'$

Examples:

- $\frac{d}{dx}[\ln(x^2 + 1)] = \frac{1}{x^2+1} \cdot 2x = \frac{2x}{x^2+1}$
- $\frac{d}{dx}[\ln(3x)] = \frac{1}{3x} \cdot 3 = \frac{1}{x}$
- $\frac{d}{dx}[\log_{10} x] = \frac{1}{x \ln 10}$

Trigonometric Functions

> **Formulas:**

> - $\frac{d}{dx}(\sin x) = \cos x$
> - $\frac{d}{dx}(\cos x) = -\sin x$
> - $\frac{d}{dx}(\tan x) = \sec^2 x$
> - $\frac{d}{dx}(\cot x) = -\csc^2 x$
> - $\frac{d}{dx}(\sec x) = \sec x \tan x$
> - $\frac{d}{dx}(\csc x) = -\csc x \cot x$

Memory aids:

- Derivatives of co-functions have negative signs
- $(\sin x)' = \cos x$
  and
  $(\cos x)' = -\sin x$

With chain rule:

- $\frac{d}{dx}(\sin u) = \cos u \cdot u'$
- $\frac{d}{dx}(\cos u) = -\sin u \cdot u'$

Examples:

- $\frac{d}{dx}[\sin(x^2)] = \cos(x^2) \cdot 2x = 2x\cos(x^2)$
- $\frac{d}{dx}[\cos(3x)] = -\sin(3x) \cdot 3 = -3\sin(3x)$
- $\frac{d}{dx}[\tan(2x)] = \sec^2(2x) \cdot 2 = 2\sec^2(2x)$

📝 Worked Examples (Ví dụ Mẫu)

Example 1: Polynomial Functions

Problem: Find the derivative of

$$f(x) = x^3 - x^2 + 2x + 1$$

Solution:

Using the sum/difference rule and power rule:

$$f'(x) = \frac{d}{dx}(x^3) - \frac{d}{dx}(x^2) + \frac{d}{dx}(2x) + \frac{d}{dx}(1)$$

$$= 3x^2 - 2x + 2 + 0$$

$$= 3x^2 - 2x + 2$$

Answer:

$$f'(x) = 3x^2 - 2x + 2$$

Example 2: Product Rule

Problem: Find the derivative of

$$y = (x^2 + 1)(x^3 - 2x)$$

Solution:

Let

$$f(x) = x^2 + 1$$

and

$$g(x) = x^3 - 2x$$

$$f'(x) = 2x$$

and

$$g'(x) = 3x^2 - 2$$

Using the product rule:

$y' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$

$= (2x)(x^3 - 2x) + (x^2 + 1)(3x^2 - 2)$

$= 2x^4 - 4x^2 + 3x^4 - 2x^2 + 3x^2 - 2$

$= 5x^4 - 3x^2 - 2$

Answer:
$y' = 5x^4 - 3x^2 - 2$

Example 3: Quotient Rule

Problem: Find the derivative of
$y = \frac{x-1}{2x+1}$

Solution:

Let
$f(x) = x - 1$
and
$g(x) = 2x + 1$

$f'(x) = 1$
and
$g'(x) = 2$

Using the quotient rule:
$y' = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$

$= \frac{(1)(2x+1) - (x-1)(2)}{(2x+1)^2}$

$= \frac{2x+1-2x+2}{(2x+1)^2}$

$= \frac{3}{(2x+1)^2}$

Answer:
$y' = \frac{3}{(2x+1)^2}$

Example 4: Chain Rule with Power

Problem: Find the derivative of
$y = (x^2 + 3x - 1)^5$

Solution:

Outer function:
$f(u) = u^5$
, so
$f'(u) = 5u^4$

Inner function:
$u = x^2 + 3x - 1$
, so
$u' = 2x + 3$

Using the chain rule:
$y' = 5(x^2 + 3x - 1)^4 \cdot (2x + 3)$

$= 5(2x + 3)(x^2 + 3x - 1)^4$

Answer:
$y' = 5(2x + 3)(x^2 + 3x - 1)^4$

Example 5: Exponential Functions

Problem: Find the derivative of
$f(x) = xe^{x^2}$

Solution:

This requires the product rule and chain rule.

Let

$u(x) = x$

and

$v(x) = e^{x^2}$

$u'(x) = 1$

For

$v'(x)$

, use chain rule:

$v'(x) = e^{x^2} \cdot 2x = 2xe^{x^2}$

Using the product rule:

$f'(x) = u'(x) \cdot v(x) + u(x) \cdot v'(x)$

$= (1)(e^{x^2}) + (x)(2xe^{x^2})$

$= e^{x^2} + 2x^2 e^{x^2}$

$= e^{x^2}(1 + 2x^2)$

Answer:

$f'(x) = e^{x^2}(1 + 2x^2)$

Example 6: Logarithmic Functions

Problem: Find the derivative of

$y = \ln(x^2 + 1)$

Solution:

Outer function:
$f(u) = \ln u$
, so
$f'(u) = \frac{1}{u}$

Inner function:
$u = x^2 + 1$
, so
$u' = 2x$

Using the chain rule:
$y' = \frac{1}{x^2+1} \cdot 2x = \frac{2x}{x^2+1}$

Answer:
$y' = \frac{2x}{x^2+1}$

Example 7: Trigonometric Functions

Problem: Find the derivative of
$f(x) = x^2 \sin x$

Solution:

Use the product rule with
$u(x) = x^2$
and
$v(x) = \sin x$

$u'(x) = 2x$
and
$v'(x) = \cos x$

$f'(x) = (2x)(\sin x) + (x^2)(\cos x)$

$= 2x\sin x + x^2 \cos x$

Answer:
$f'(x) = 2x\sin x + x^2 \cos x$

Example 8: Combined Rules

Problem: Find the derivative of
$y = \frac{e^x}{x^2+1}$

Solution:

Use the quotient rule with
$f(x) = e^x$
and
$g(x) = x^2 + 1$

$f'(x) = e^x$
and
$g'(x) = 2x$

$$y' = \frac{(e^x)(x^2+1)-(e^x)(2x)}{(x^2+1)^2}$$

$$= \frac{e^x(x^2+1-2x)}{(x^2+1)^2}$$

$$= \frac{e^x(x^2-2x+1)}{(x^2+1)^2}$$

$$= \frac{e^x(x-1)^2}{(x^2+1)^2}$$

Answer:
$y' = \frac{e^x(x-1)^2}{(x^2+1)^2}$

🌍 Applications (Ứng dụng Thực tế)

Application 1: Velocity and Acceleration

Problem: An object is thrown vertically upward with initial velocity $v_0 = 20$ m/s. Its height (in meters) after t seconds is given by:

$$h(t) = v_0 t - \frac{1}{2} g t^2 = 20t - 4.9t^2$$

where $g = 9.8$ m/s² is gravitational acceleration.

a) Find the velocity function $v(t)$.

b) Find the acceleration function $a(t)$.

c) At what time does the object reach its maximum height?

d) What is the velocity when the object hits the ground?

Solution:

a) Velocity is the derivative of position:

$$v(t) = h'(t) = 20 - 9.8t$$

b) Acceleration is the derivative of velocity:

$a(t) = v'(t) = -9.8$ m/s²

(This is constant, equal to $-g$, as expected)

c) Maximum height occurs when $v(t) = 0$:

$$20 - 9.8t = 0$$

$t = \frac{20}{9.8} \approx 2.04$
seconds

d) The object hits the ground when
$h(t) = 0$
:
$20t - 4.9t^2 = 0$
$t(20 - 4.9t) = 0$
$t = 0$
(initial) or
$t = \frac{20}{4.9} \approx 4.08$
seconds

At
$t = 4.08$
:
$v(4.08) = 20 - 9.8(4.08) = 20 - 40 = -20$
m/s

Answer: The object hits the ground with velocity
-20
m/s (negative indicates downward direction).

Application 2: Marginal Cost in Economics

Problem: A company's total cost (in millions of dollars) to produce
x
thousand units of a product is:
$C(x) = 0.001x^3 - 0.3x^2 + 40x + 1000$

a) Find the marginal cost function
$C'(x)$
.
b) Calculate the marginal cost when producing 100,000 units (
$x = 100$
).
c) Interpret the result.

Solution:

a) The marginal cost is the derivative:

$C'(x) = 0.003x^2 - 0.6x + 40$

b) At
$x = 100$
:
$C'(100) = 0.003(100)^2 - 0.6(100) + 40$
$= 0.003(10{,}000) - 60 + 40$
$= 30 - 60 + 40 = 10$

c) **Interpretation:** When producing 100,000 units, the marginal cost is $10 million per thousand units, or $10,000 per unit. This means producing one additional unit (the 100,001st unit) will cost approximately $10,000.

✏️ Practice Exercises (Bài tập Luyện tập)

Exercise 1: Find the derivatives of the following functions:

a)
$f(x) = 5x^4 - 3x^2 + 7x - 2$
b)
$g(x) = \sqrt{x} + \frac{1}{x^2}$
c)
$h(x) = (x^2 + 1)(x^3 - x)$
d)
$k(x) = \frac{x^2 - 1}{x + 2}$

Exercise 2: Use the chain rule to find the derivatives:

a)
$f(x) = (2x^3 - 5x)^{10}$
b)
$g(x) = \sqrt{x^2 + 4}$
c)
$h(x) = e^{x^2 + 3x}$
d)
$k(x) = \ln(x^3 - 2x + 1)$

Exercise 3: Find the derivatives of trigonometric functions:

a)
$f(x) = \sin(3x)$

b)
$g(x) = \cos^2 x$
(Hint:
$\cos^2 x = (\cos x)^2$
)

c)
$h(x) = x \sin x$

d)
$k(x) = \frac{\sin x}{x}$

Exercise 4: Combined rules:

a)
$f(x) = xe^{-x}$

b)
$g(x) = \frac{\ln x}{x}$

c)
$h(x) = e^x \sin x$

📋 Review Exercises (Bài tập Ôn tập)

Exercise 1: Compute the following derivatives:

a)
$\frac{d}{dx}[(x^2 + 1)^3(x - 2)^2]$

b)
$\frac{d}{dx}\left[\frac{e^x}{1+e^x}\right]$

c)
$\frac{d}{dx}[\ln(\sin x)]$

d)
$\frac{d}{dx}[\sin(e^{x^2})]$

Exercise 2: Find $f'(x)$ for:

a) $f(x) = (x^3 - 2x)^5$

b) $f(x) = \sqrt{x^2 + \sqrt{x}}$

c) $f(x) = e^{x^2} \ln x$

Exercise 3: Application problems

a) The position of a particle moving along a line is $s(t) = t^3 - 6t^2 + 9t$ meters at time t seconds. Find the velocity and acceleration at $t = 2$ seconds.

b) The temperature T (in °C) of a cooling object after t minutes is $T(t) = 25 + 70e^{-0.5t}$. Find the rate of cooling at $t = 5$ minutes.

Exercise 4: Tangent line equations

Find the equation of the tangent line to the curve at the given point:

a) $y = x^3 - 2x + 1$ at $x = 1$

b)
$y = \frac{1}{x}$
at
$x = 2$
c)
$y = e^x$
at
$x = 0$

📝 Chapter Summary (Tóm tắt Chương)

Key Derivative Rules

Rule	Formula	Example
Constant	$(c)' = 0$	$(5)' = 0$
Power	$(x^n)' = nx^{n-1}$	$(x^3)' = 3x^2$
Constant Multiple	$(cf)' = cf'$	$(3x^2)' = 6x$
Sum	$(f+g)' = f' + g'$	$(x^2 + x)' = 2x + 1$
Difference	$(f-g)' = f' - g'$	$(x^3 - x)' = 3x^2 - 1$
Product	$(fg)' = f'g + fg'$	$(x \cdot e^x)' = e^x + xe^x$
Quotient	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$	$\left(\frac{x}{x+1}\right)' = \frac{1}{(x+1)^2}$
Chain	$(f \circ g)' = f'(g) \cdot g'$	$[(x^2)^3]' = 3(x^2)^2 \cdot 2x$

Derivatives of Common Functions

Function	Derivative
x^n	nx^{n-1}
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$\frac{1}{x}$
$\log_a x$	$\frac{1}{x \ln a}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$

Problem-Solving Strategies

1. **Identify the type of function:** polynomial, rational, exponential, logarithmic, trigonometric, or composite
2. **Choose the appropriate rule(s):** power, product, quotient, chain
3. **Apply rules systematically:** work from outside to inside for composite functions
4. **Simplify when possible:** before or after differentiation
5. **Check your work:** verify dimensions and special values

Common Mistakes to Avoid

❌
$(fg)' eq f' g'$

— Use product rule:
$(fg)' = f'g + fg'$
❌
$\left(\frac{f}{g}\right)' eq \frac{f'}{g'}$
— Use quotient rule
❌
$(f \circ g)' eq f' \circ g'$
— Use chain rule:
$(f \circ g)' = f'(g) \cdot g'$
❌ Forgetting the chain rule when differentiating composite functions
❌ Sign errors in quotient rule (remember:
$f'g - fg'$
, not
$fg' - f'g$
)

🎯 **Looking Ahead:** In the next chapter, we will explore **applications of derivatives**, including finding maximum and minimum values, analyzing the shape of curves, solving optimization problems, and using derivatives to sketch accurate graphs of functions. These techniques are essential for solving real-world problems in engineering, economics, and the sciences.